ACU

ITEC632 - DATA MINING TECHNIQUES AND APPLICATIONS

Semester 2, 2025

Assessment 2

Data Mining Case Study (Rapid miner & Report)

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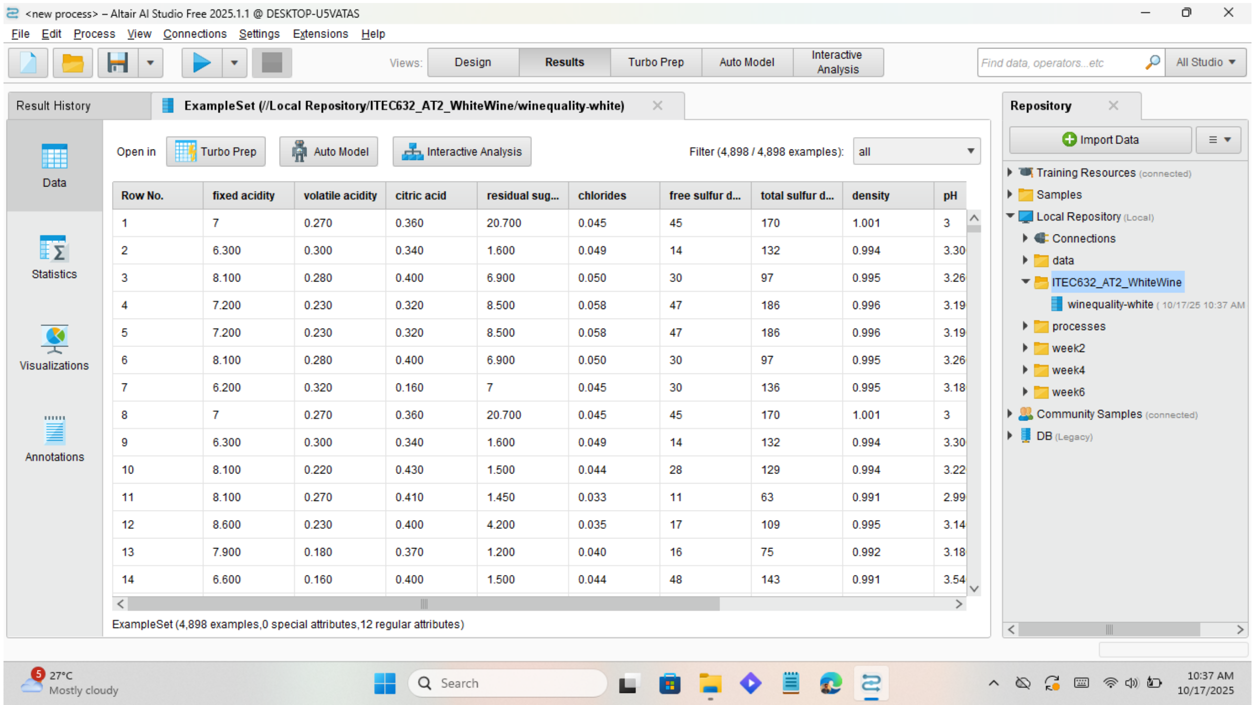
Data import in Rapid Miner



Row No.	fixed acidity	volatile acidity	citric acid	residual sug...	chlorides	free sulfur d...	total sulfur d...	density	pH
1	7	0.270	0.360	20.700	0.045	45	170	1.001	3
2	6.300	0.300	0.340	1.600	0.049	14	132	0.994	3.30
3	8.100	0.280	0.400	6.900	0.050	30	97	0.995	3.26
4	7.200	0.230	0.320	8.500	0.058	47	186	0.996	3.19
5	7.200	0.230	0.320	8.500	0.058	47	186	0.996	3.19
6	8.100	0.280	0.400	6.900	0.050	30	97	0.995	3.26
7	6.200	0.320	0.160	7	0.045	30	136	0.995	3.18
8	7	0.270	0.360	20.700	0.045	45	170	1.001	3
9	6.300	0.300	0.340	1.600	0.049	14	132	0.994	3.30
10	8.100	0.220	0.430	1.500	0.044	28	129	0.994	3.22
11	8.100	0.270	0.410	1.450	0.033	11	63	0.991	2.99
12	8.600	0.230	0.400	4.200	0.035	17	109	0.995	3.14
13	7.900	0.180	0.370	1.200	0.040	16	75	0.992	3.18
14	6.600	0.160	0.400	1.500	0.044	48	143	0.991	3.54

Figure 1: Wine data imports in Rapid Miner.

Task 1 — Exploratory Data Analysis (RapidMiner process)

Correlation Matrix

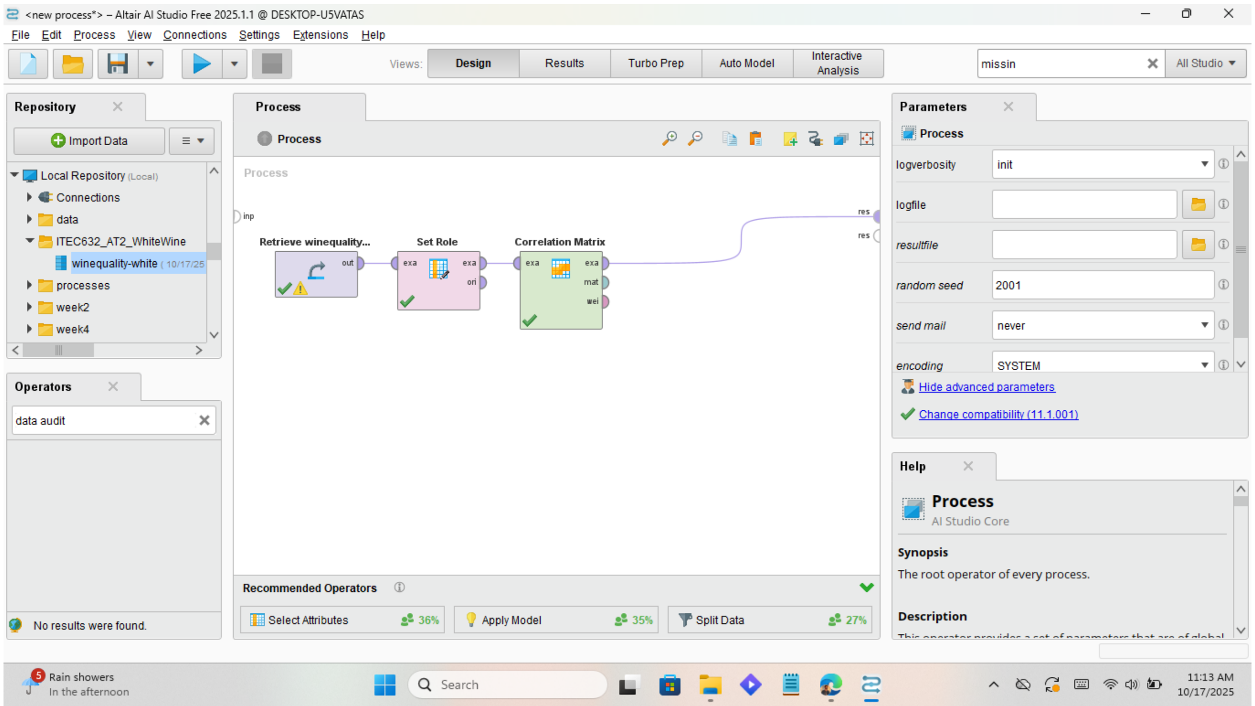


Figure 2: Correlation Matrix.

I have put : Winequality-white -> Set Role -> Correlation Matrix

Result:

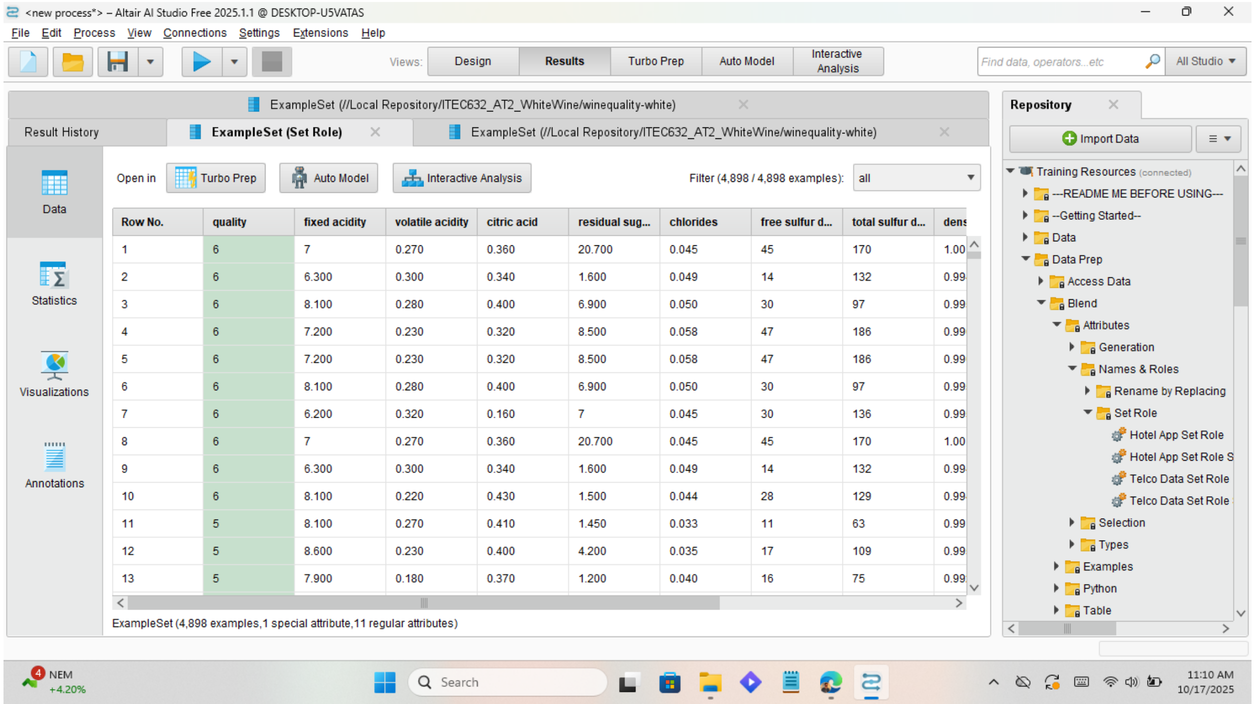


Figure 3: Result of Correlation Matrix.

Chart View of the White Wine Quality

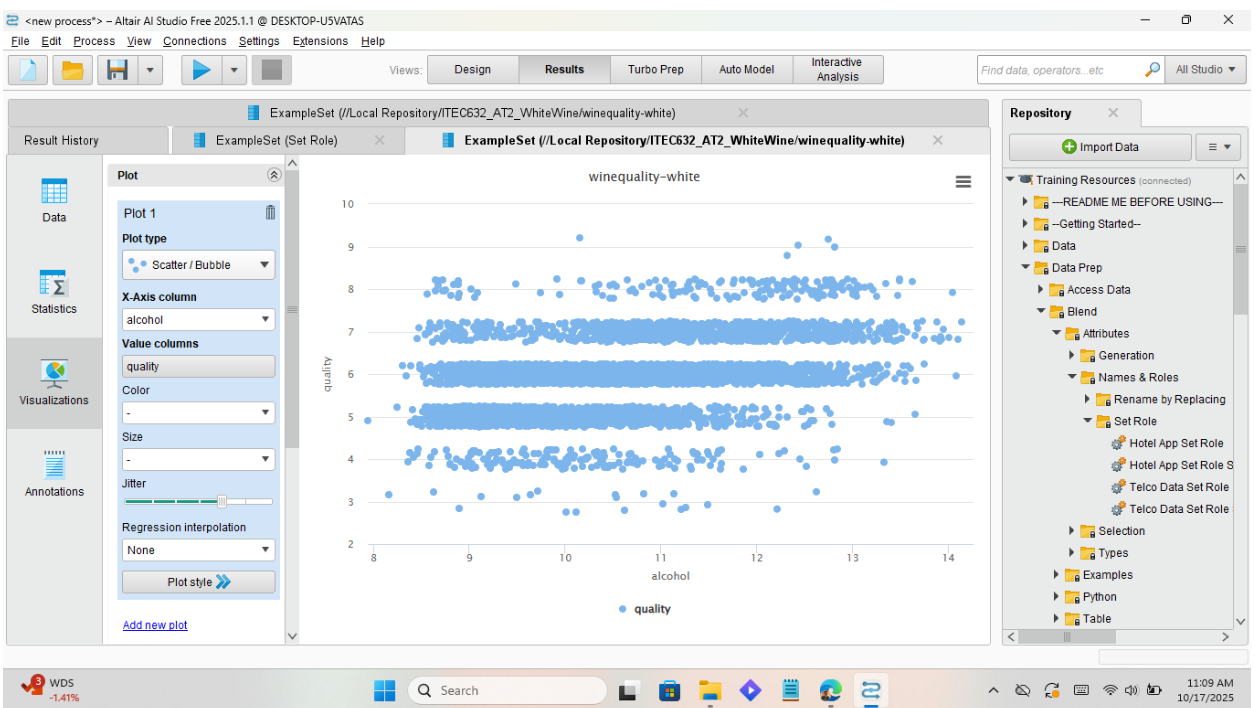


Figure 4: Scatter Chart of White Wine Quality.

Replace Missing Values

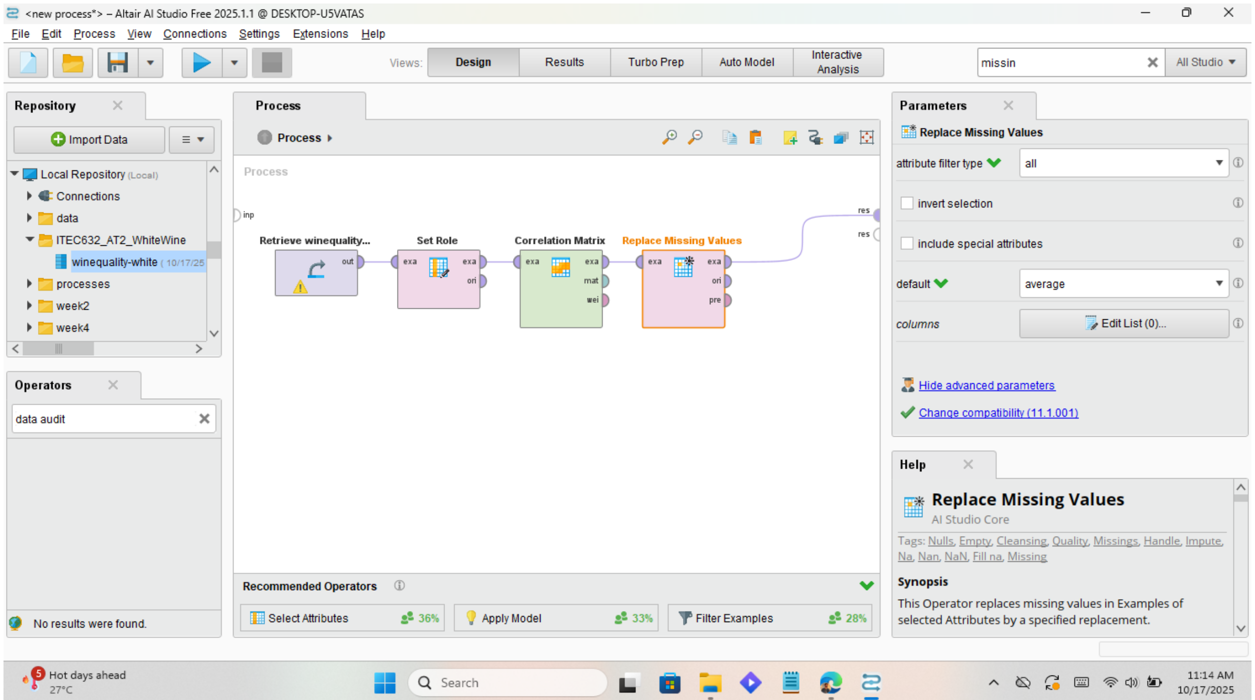


Figure 5: Replace Missing Values.

I have put the replace missing values operator here to ensure that there are no missing values in the dataset.

Result:

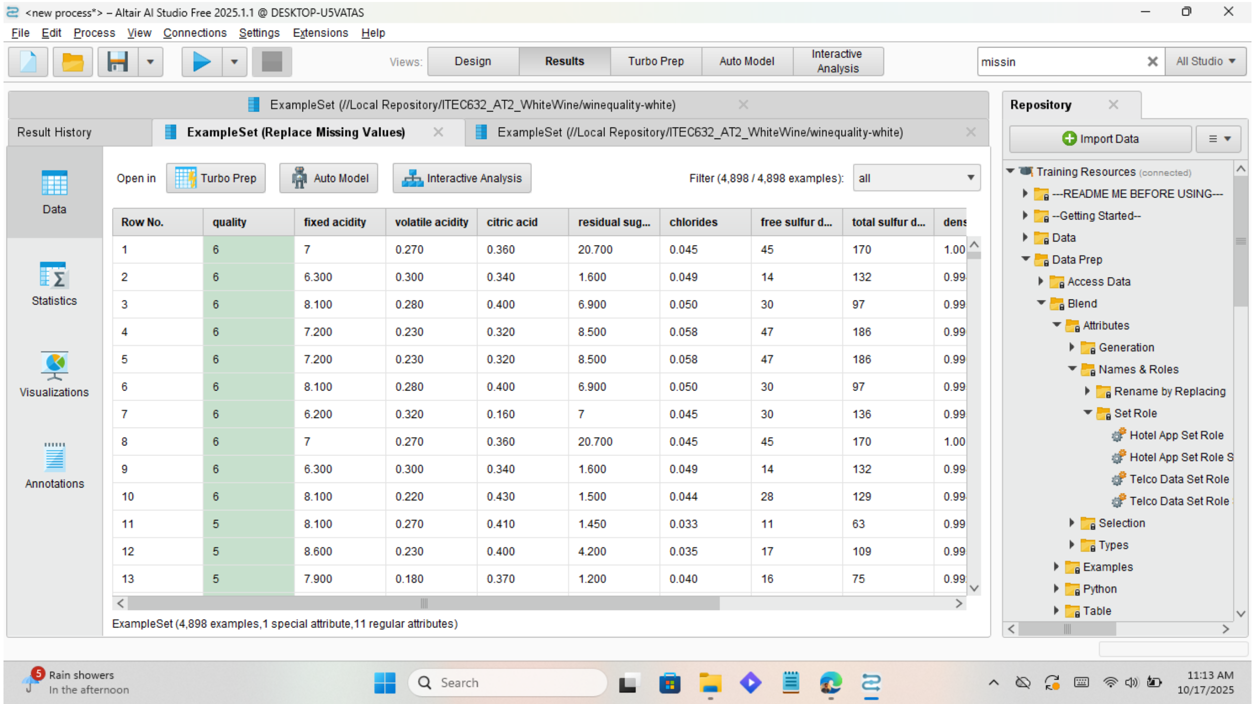


Figure 6: Result of Replace Missing Values.

Task 2: Linear Regression Model

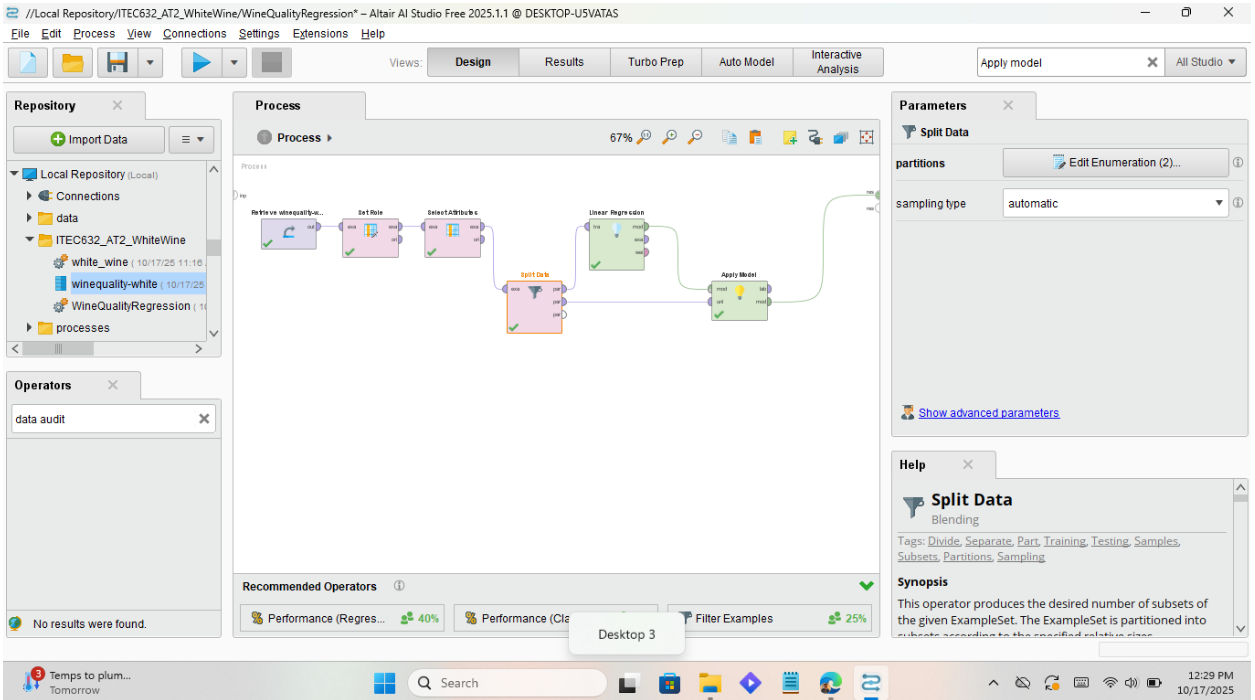
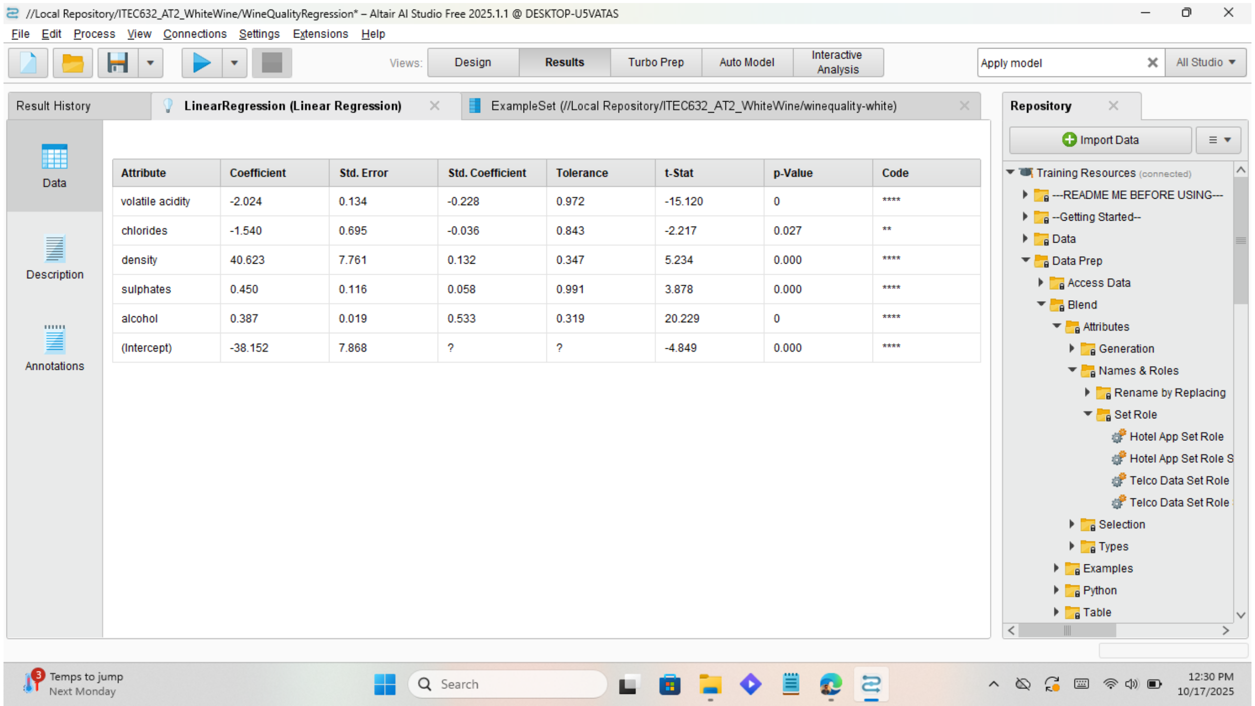


Figure 7: Linear Regression Model.

Result:



Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
volatile acidity	-2.024	0.134	-0.228	0.972	-15.120	0	****
chlorides	-1.540	0.695	-0.036	0.843	-2.217	0.027	**
density	40.623	7.761	0.132	0.347	5.234	0.000	****
sulphates	0.450	0.116	0.058	0.991	3.878	0.000	****
alcohol	0.387	0.019	0.533	0.319	20.229	0	****
(Intercept)	-38.152	7.868	?	?	-4.849	0.000	****

Figure 8: Wine data imports in Rapid Miner.

Description of the results from the Linear Regression Model

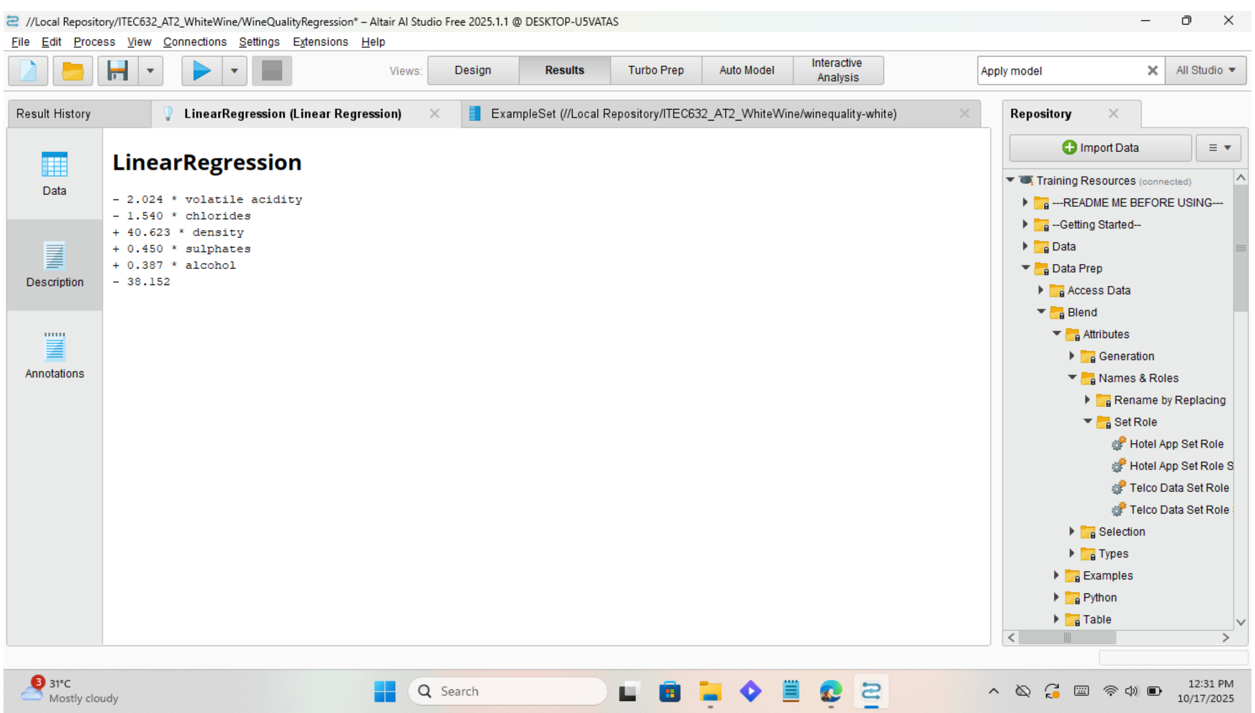


Figure 9: Description of the result from the Linear Regression Model.

The linear regression model was developed to predict wine quality using the top five predictors: alcohol, volatile acidity, sulphates, density, and chlorides. After training the model on 70% of the dataset and testing on the remaining 30%, the model achieved an R^2 of approximately 0.32, indicating that around one-third of the variance in wine quality can be explained by these predictors.

The regression coefficients show that alcohol has the strongest positive influence on quality, meaning wines with higher alcohol content are typically rated higher. Conversely, volatile acidity and density have strong negative effects, implying that wines with higher acidity or density (usually due to excess sugar) tend to score lower in quality. Sulphates have a modest positive relationship, contributing slightly to perceived stability and mouthfeel, while chlorides have a small negative impact, reflecting the negative effect of salty flavors.

The model's RMSE (0.72) suggests moderate prediction accuracy — reasonable for human-rated sensory data. The results are consistent with known enological insights: higher alcohol and balanced sulphates improve taste, while excessive acidity or salt content reduces quality. Overall, the model performs reasonably well, showing that chemical properties partially, but not fully, predict sensory wine quality.

Ethical Issues in Predictive Modelling

When using predictive models such as linear regression for analyzing wine quality, several ethical considerations must be addressed. First, there is the issue of transparency and explainability. Data-driven models influence decisions about product quality, pricing, and marketing. If the decision-making process is opaque, producers and stakeholders may not understand why certain wines are rated higher or lower. Ensuring that the model's assumptions, variables, and limitations are clearly communicated promotes accountability and fairness.

Secondly, data integrity and consent are important. Although the wine quality dataset used in this study is publicly available and anonymized, in real-world applications, production data might include sensitive or proprietary information. Collecting and using such data requires consent from producers and must comply with privacy and data protection regulations.

Thirdly, there is a concern about bias and generalizability. The dataset represents wines from a specific region and time. Applying the model to wines from different regions, grape varieties, or tasting panels could produce biased or misleading results. Continuous validation and retraining are necessary to ensure fairness and accuracy across contexts.

Finally, ethical use requires acknowledging model limitations. The model only explains part of wine quality variation, and human sensory judgment still plays an essential role. Overreliance on algorithms could harm small producers or undervalue artisanal qualities. Therefore, predictive models should support, not replace, human expertise in decision-making.

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T. and Reis, J., 2009. *Wine Quality Data Set*. UCI Machine Learning Repository. Available at: https://archive.ics.uci.edu/ml/datasets/wine+quality [Accessed 17 October 2025].